

Comparison of Chest Radiography, Clinical, and Late Fusion Models for Predicting In-Hospital Mortality in Patients With COVID-19

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Key Revisions After Faculty Feedback

Faculty feedback / Previous issue

Revision

Prediction time point was unclear



Defined T0 as visit_start_datetime

CXR selection could include information after admission



Reconstructed CXR selection around T0

Too many clinical predictors for the number of events



131 variables audited → 14 candidates → 9 final raw variables

Clinical predictors required stronger clinical and literature-based justification



Selected clinically relevant candidates and performed backward elimination using AIC

Missing-value imputation lacked a clear rationale



Assessed distributions using the Training set and selected preprocessing accordingly

CXR model selected a very early checkpoint



Re-evaluated learning rate, augmentation, class weighting, and learning curves

Model hyperparameters were not clearly reported



Explicitly documented architecture, training conditions, hyperparameters, and model-selection criteria

Fusion strategy required reconsideration



Changed Early Fusion → Late Fusion

Contribution of each modality in Late Fusion was unclear



Evaluated modality-level permutation importance for Clinical and CXR predictions

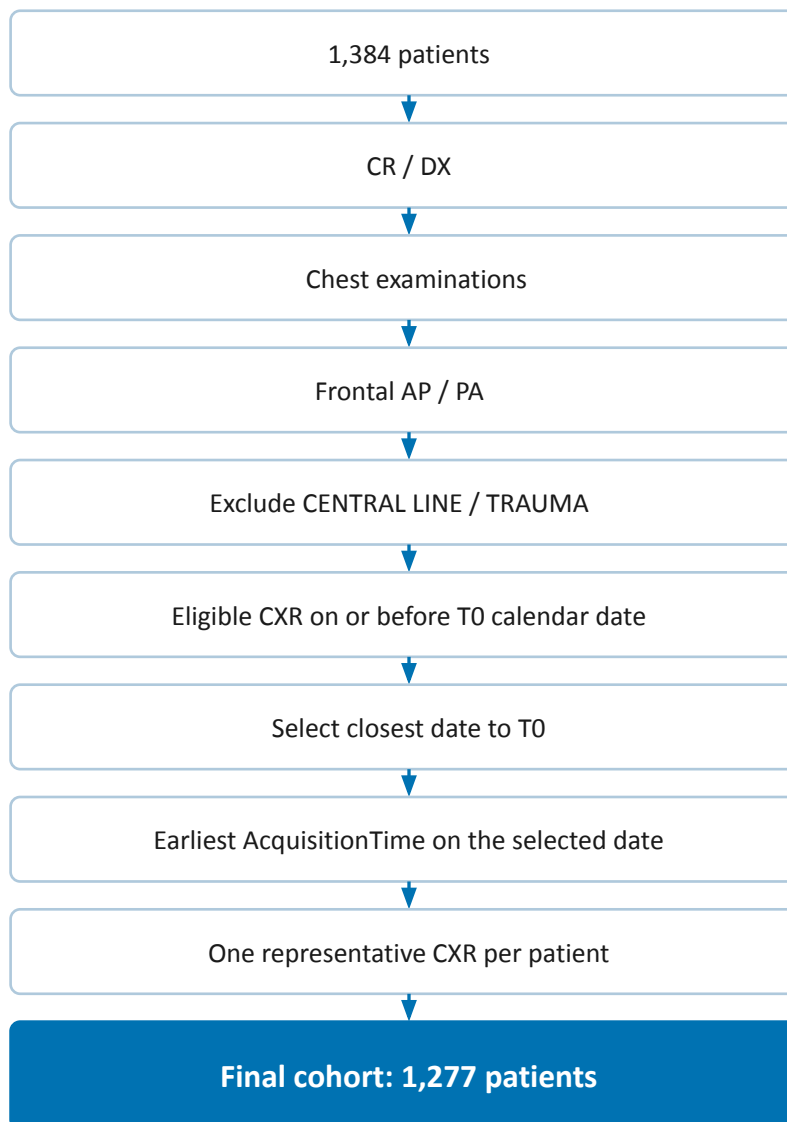
Prediction Time Point and CXR Selection

Previous issue

- Prediction time point was not clearly defined
- CXR selection could include post-admission images
- This could introduce future information

T0 = visit_start_datetime

Limitation: exact admission time was unavailable; therefore, CXRs obtained on the T0 calendar date could not be strictly classified as pre- or post-admission.



Why this window?

Among 1,309 patients with an eligible CXR on or before the T0 calendar date

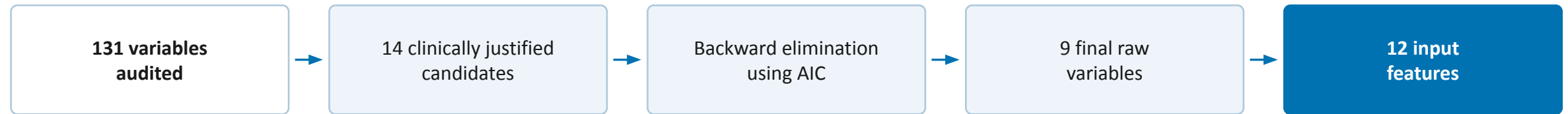
97.6%

1,277 patients had their nearest CXR within T0-2 days through the T0 calendar date

Final inclusion window
T0-2 days → T0 calendar date

CXRs obtained on calendar dates after T0 were not used, to avoid future information reflecting post-admission progression or treatment.

Clinical Variable Selection and Missing Data Handling



Nine final raw variables

SpO ₂	Lymphocyte count	eGFR
CRP	D-dimer	Lactate
Age	Sex	Heart failure

Why these candidates?

- Selected for clinical relevance to COVID-19 in-hospital mortality
- Supported by previous evidence and established risk scores
- Training set contained 135 deaths — about 13–14 raw variables set as a practical upper limit to avoid too many predictors for the number of events

Missing data and preprocessing

- Distribution assessment: Training set only
- Descriptive statistics + Shapiro–Wilk test
- Continuous variables showed non-normality and/or asymmetry
- Missing continuous values → Training-set median imputation
- Selected skewed variables → log1p transformation
- Same fitted preprocessing applied unchanged to Validation and Test sets
- No automatic exclusion of statistical outliers

Model Rebuilding, Hyperparameters, and Late Fusion

CXR ResNet18

ImageNet-pretrained ResNet18, full fine-tuning

Re-evaluated systematically

- 2 augmentation strategies
- 3 learning rates
- With vs without class weighting
- Learning-curve stability

Previously the best checkpoint fell at a very early epoch.

Final setting

Augmentation B: mild translation/scale, brightness/contrast, $\pm 5^\circ$ rotation, no flip

Learning rate = 1×10^{-5}

Batch size = 32

BCEWithLogitsLoss, pos_weight = 6.56

AdamW, weight decay = 1×10^{-4}

Best checkpoint = epoch 12

Clinical models

All three models used the same 12 input features

Model	Key final hyperparameters
Logistic Regression	C = 1.0, class_weight = None, L2, liblinear
XGBoost	max_depth = 3, learning_rate = 0.10, min_child_weight = 1, scale_pos_weight = 1.0, best iteration = 24
MLP	12 inputs, 64→32, dropout = 0.2, learning rate = 1×10^{-4} , batch size = 64, best epoch = 24

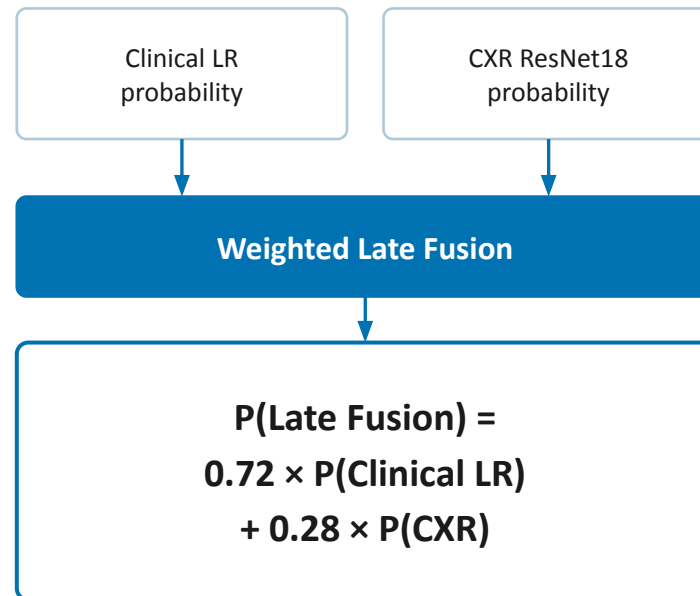
Model and hyperparameter selection

Training + Validation only

The fixed Test set was not used for any model or hyperparameter choice.

Late Fusion

Early Fusion → **decision-level Late Fusion**



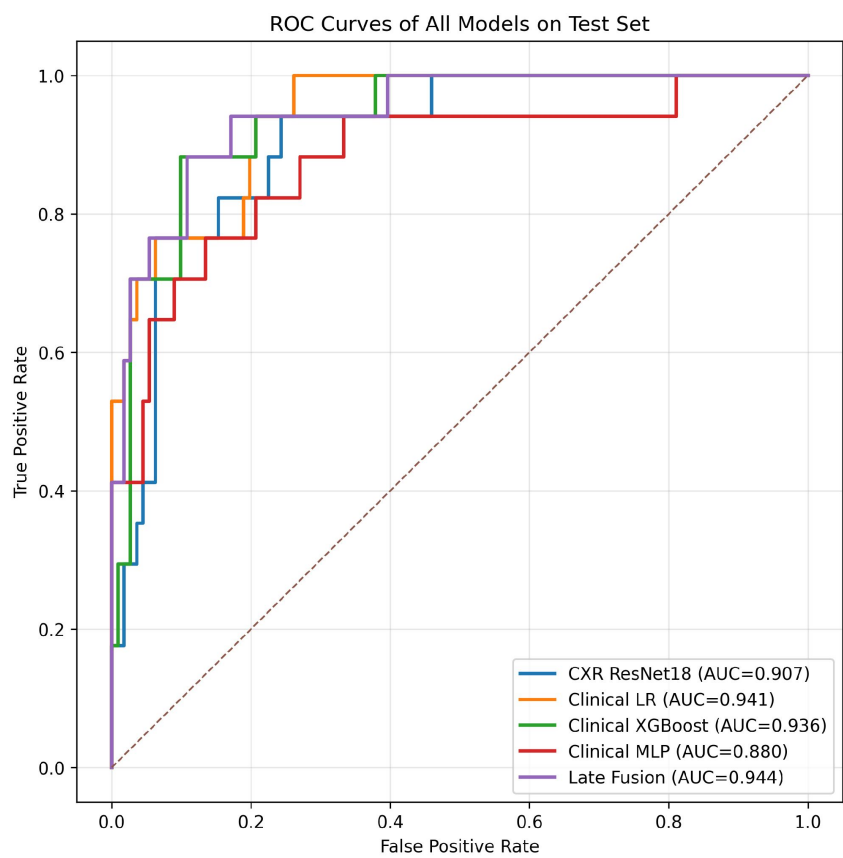
- Fusion weights selected using the Validation set only
- Clinical LR weight = 0.72, CXR weight = 0.28

Test set remained locked until final evaluation

Final Model Performance

Fixed Test set: n = 128, deaths = 17

1 ROC curves — five models



2 Performance on the fixed Test set — five models

Model	ROC-AUC	95% CI	PR-AUC	Brier score	ECE
CXR ResNet18	0.907	0.836–0.963	0.623	0.102	0.152
Clinical Logistic Regression	0.941	0.887–0.981	0.800	0.057	0.041
XGBoost	0.936	0.876–0.979	0.718	0.068	0.090
MLP	0.880	0.768–0.961	0.689	0.169	0.309
Late Fusion	0.944	0.884–0.985	0.797	0.059	0.065

3 Paired DeLong test with Holm correction

Late Fusion vs Clinical LR

+0.003

ΔROC-AUC

0.950

Holm-adjusted P

Not significant

Clinical LR also showed the best calibration-related performance — **highest PR-AUC, lowest Brier score and ECE** among the five models.

Late Fusion had the highest ROC-AUC point estimate, but did not significantly outperform Clinical LR.

Late Fusion Modality Importance

Modality-level permutation importance on the fixed Test set (1,000 permutations, seed 42)

0.944

Baseline Late Fusion
ROC-AUC

0.169

Mean Δ AUC after permuting
Clinical probability

0.045

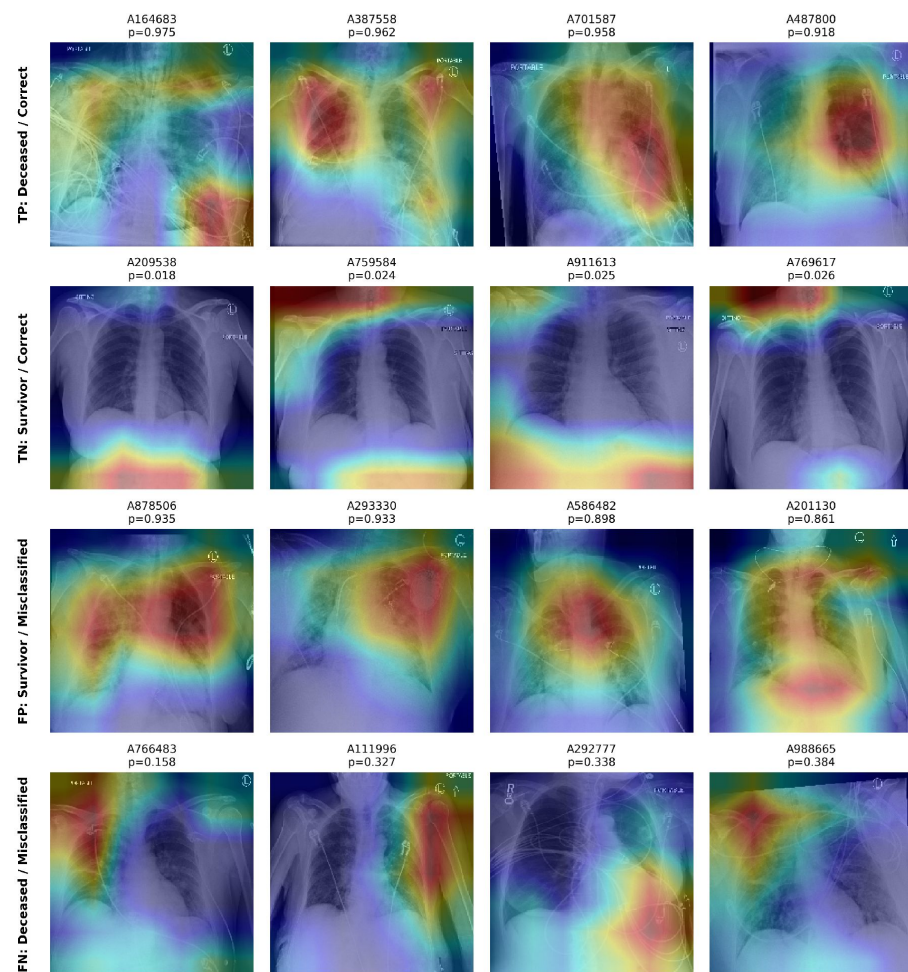
Mean Δ AUC after permuting
CXR probability

Late Fusion relied predominantly on clinical information, while CXR provided complementary information.

This analysis describes the model's dependence on each modality; it was not used to modify the model specification or to retune the fusion weights.

Grad-CAM of Representative CXR Predictions

Grad-CAM for In-Hospital Mortality Prediction



True positive

Activation mainly within the lung fields, partially overlapping radiographic infiltrates.

True negative

Limited mortality-class activation within the lung fields; relative activation often outside the lungs.

False positive

Prominent activation within lung regions showing extensive infiltrates.

False negative

Limited activation within the lung fields, with activation also in extrapulmonary regions such as bony structures, image margins, and medical devices.

Grad-CAM is an explanatory aid; it does not provide a diagnosis or establish causal relationships.

Conclusion

- 1 The study design was rebuilt to better align predictors with the prediction time point and reduce the risk of information leakage.
- 2 Clinical Logistic Regression achieved high discrimination and the best calibration-related performance.
- 3 Late Fusion achieved the highest ROC-AUC point estimate, but did not significantly outperform Clinical LR.

The incremental predictive value of adding CXR was limited, and external validation is required.

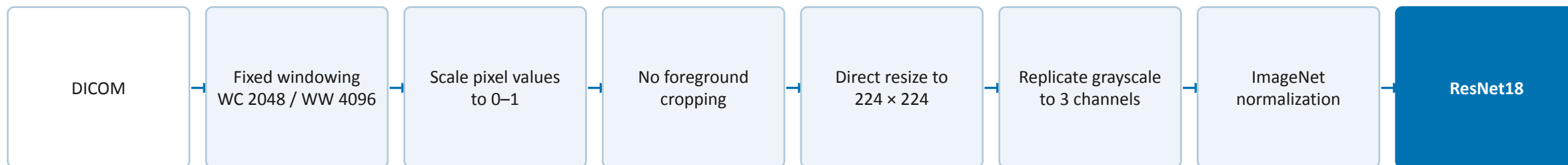
Supplementary Slides

For discussion

Detailed CXR Selection Criteria

Criterion	Definition
DICOM modality	CR / DX
Chest examination	Identified from Study Description
Projection	Frontal AP / PA (from Series Description)
Time window	T0–2 days through the T0 calendar date
Excluded study purpose	CENTRAL LINE confirmation; TRAUMA examination
Not excluded by name alone	CODE T; INFANT; OVER PENETRATED
Representative date	Acquisition date closest to T0
Same-date tie-breaker 1	Earliest AcquisitionTime
Same-date tie-breaker 2	Smallest SeriesNumber only when AcquisitionTime was tied
Images retained per patient	1
Final cohort	1,277 patients / 1,277 CXRs

CXR Image Preprocessing



DICOM header audit (all 1,277 images)

- Photometric Interpretation: MONOCHROME2
- Rescale Slope = 1.0, Rescale Intercept = 0.0
- No padding to preserve the original aspect ratio
- Validation / Test: no random augmentation

Training augmentation B

- Translation $\pm 2\%$, scale 0.98–1.02
- Brightness 0.95–1.05, contrast 0.95–1.05
- Rotation $\pm 5^\circ$
- No horizontal or vertical flip

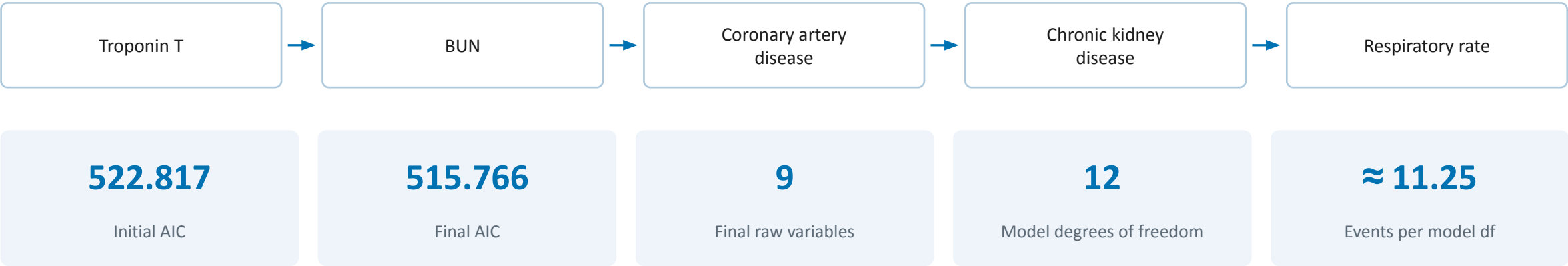
Clinical Variable Selection Details

14 candidate variables

SpO₂	Respiratory rate	Lymphocyte count	eGFR	CRP
D-dimer	Lactate	BUN	Troponin T	Age
Sex	Heart failure	Chronic kidney disease	Coronary artery disease	

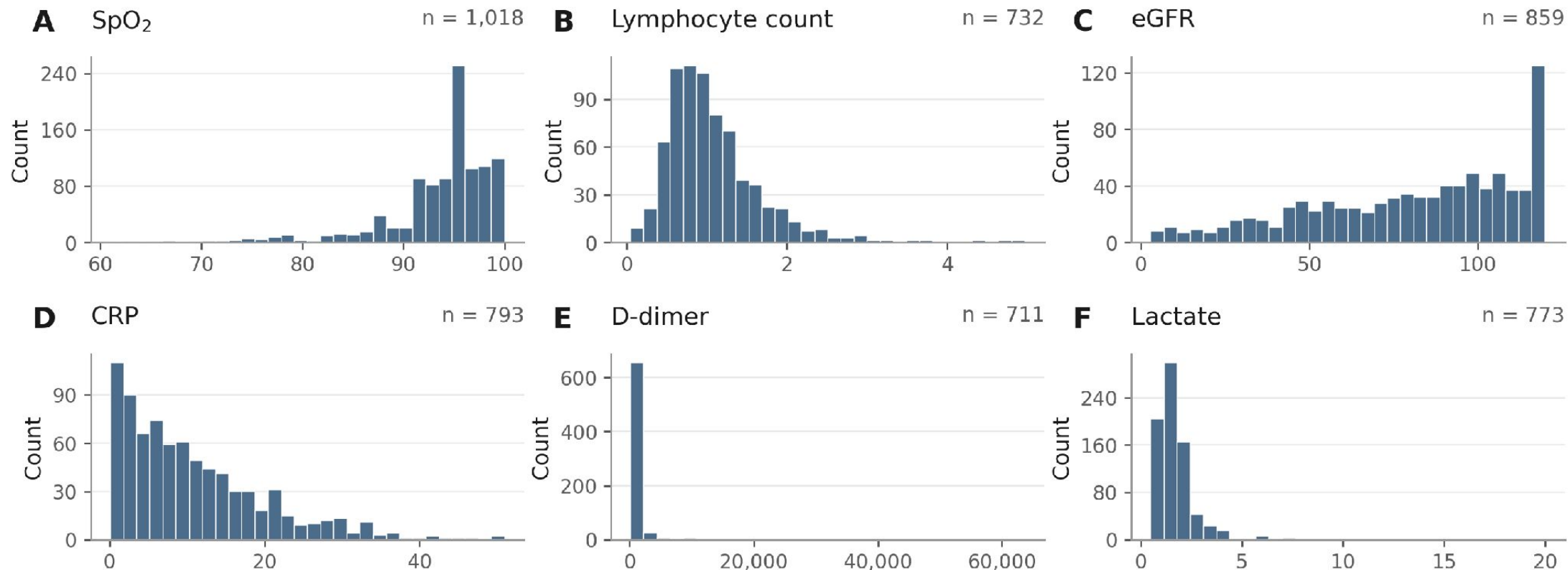
Shaded = removed by backward elimination Bold = retained in the final model

Backward AIC removal order



Raw Distributions of Continuous Clinical Variables in the Training Set

Training set only (n = 1,021) — raw observed values



Missing values were excluded from visualization; n above each panel is the number of observed values.

No imputation, log1p transformation, or standardization was applied to these plots.

Preprocessing decisions were based on the raw Training-set distributions without reference to the Validation or Test sets.

Clinical Preprocessing Details

Feature	Missing-value handling	Transformation / encoding	Reference or scaling
SpO ₂	Training-set median	None	Standardized (Training mean/SD)
Lymphocyte count	Training-set median	log1p	Standardized (Training mean/SD)
eGFR	Training-set median	None	Standardized (Training mean/SD)
CRP	Training-set median	log1p	Standardized (Training mean/SD)
D-dimer	Training-set median	log1p	Standardized (Training mean/SD)
Lactate	Training-set median	log1p	Standardized (Training mean/SD)
Age	Not applicable	60–74 and 75–90 indicators	18–59 years reference
Sex	Training-set mode (male)	Male indicator	Female reference
Heart failure	Missing retained as category	HFpEF, HFrEF, missing indicators	No heart failure reference

Final input features: 12 — the same features were used for Logistic Regression, XGBoost, and MLP.

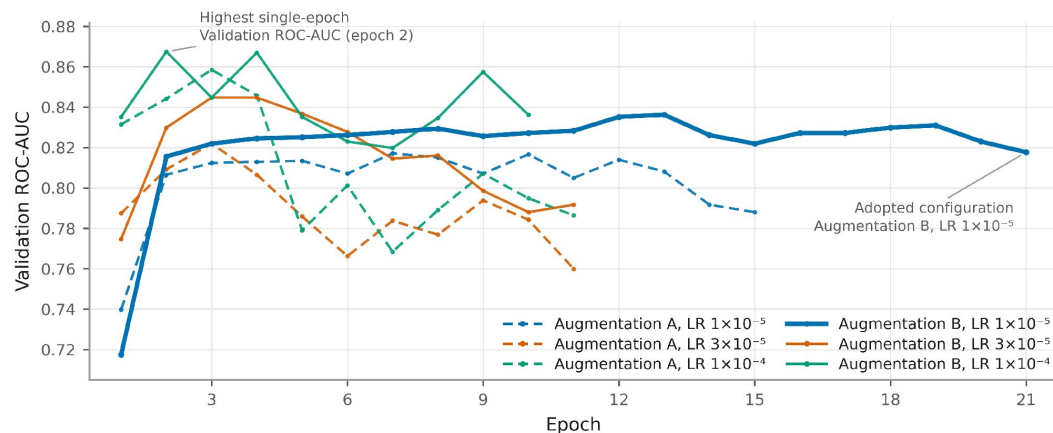
Model Hyperparameters

Model	Search / comparison	Final selected condition	Validation performance
Clinical Logistic Regression	C = {0.01, 0.1, 1, 10}; class_weight = {None, balanced}; L2; liblinear; max_iter = 5,000; seed = 42	C = 1.0; class_weight = None	ROC-AUC 0.896; PR-AUC 0.601; Brier 0.079; ECE 0.076
Clinical XGBoost	Stage 1: max_depth {2, 3, 4}, learning_rate {0.03, 0.05, 0.10}, min_child_weight {1, 5} Stage 2: subsample, colsample_bytree, scale_pos_weight, reg_alpha	max_depth = 3; learning_rate = 0.10; min_child_weight = 1; subsample = 0.8; colsample_bytree = 0.8; scale_pos_weight = 1.0; reg_alpha = 0; reg_lambda = 1; best iteration = 24	ROC-AUC 0.886; PR-AUC 0.539; Brier 0.086; ECE 0.064
Clinical MLP	Stage 1: hidden layers {64→32, 32→16}; dropout {0.2, 0.3}; learning rate {1×10 ⁻⁴ , 3×10 ⁻⁴ , 1×10 ⁻³ } Limited reevaluation: batch size {32, 64}; weight decay {1×10 ⁻⁴ , 1×10 ⁻³ }	12 inputs; 64→32; BatchNorm1d; ReLU; dropout 0.2; BCEWithLogitsLoss; AdamW; lr 1×10 ⁻⁴ ; weight decay 1×10 ⁻⁴ ; batch 64; pos_weight 1.0; max 200 epochs; patience 20; best epoch 24	ROC-AUC 0.918; PR-AUC 0.718
CXR ResNet18	2 augmentation strategies × 3 initial learning rates (1×10 ⁻⁵ , 3×10 ⁻⁵ , 1×10 ⁻⁴); with vs without class weighting	Augmentation B; lr 1×10 ⁻⁵ ; BCEWithLogitsLoss pos_weight 6.56; AdamW; weight decay 1×10 ⁻⁴ ; batch 32; max 30 epochs; ReduceLROnPlateau (factor 0.5, patience 3); early stopping patience 8; best epoch 12	ROC-AUC 0.849; PR-AUC 0.561

All searches used the Training and Validation sets only; the Test set was not used for hyperparameter or model selection.

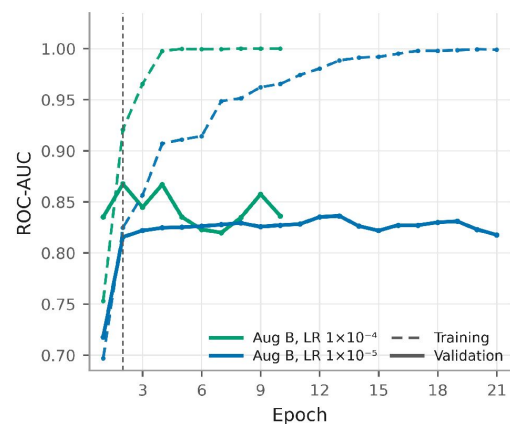
CXR Training Configuration Selection

A Validation ROC-AUC, six candidate configurations

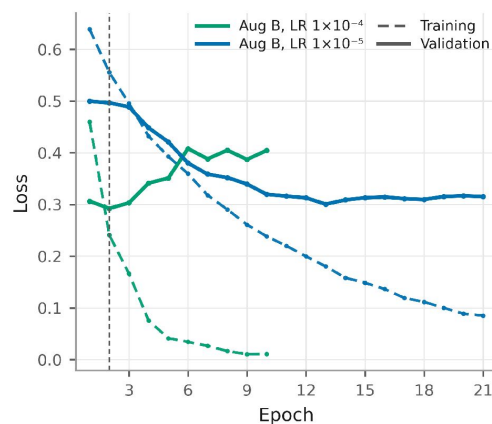


- 2 augmentation strategies $\times$ 3 initial learning rates, without class weighting
- Augmentation B with 1×10^{-4} gave the highest single-epoch Validation ROC-AUC, but its best epoch was 2
- Training ROC-AUC then rose rapidly while Validation loss increased and Validation ROC-AUC declined
- Learning curves were therefore reviewed before fixing the configuration

B Training and Validation ROC-AUC



C Training and Validation loss



Final selection

Augmentation B + learning rate 1×10^{-5} + class weighting (pos_weight 6.56)

Final best epoch: 12

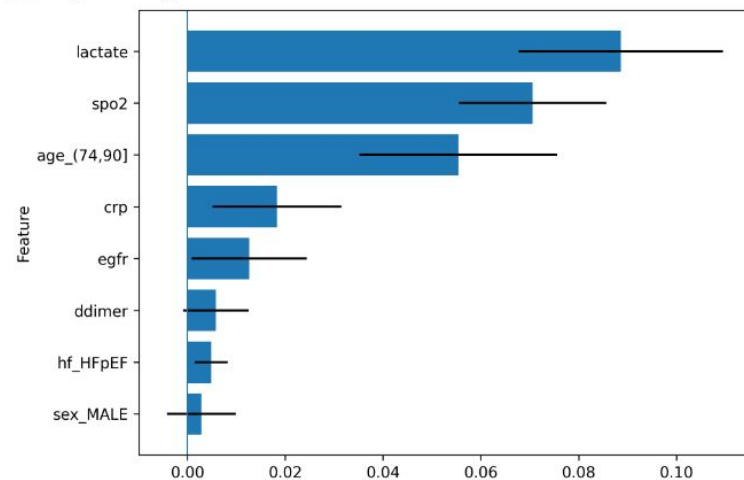
With class weighting, Validation ROC-AUC 0.849 and PR-AUC 0.561, compared with 0.836 and 0.540 without class weighting.

Clinical Feature Importance

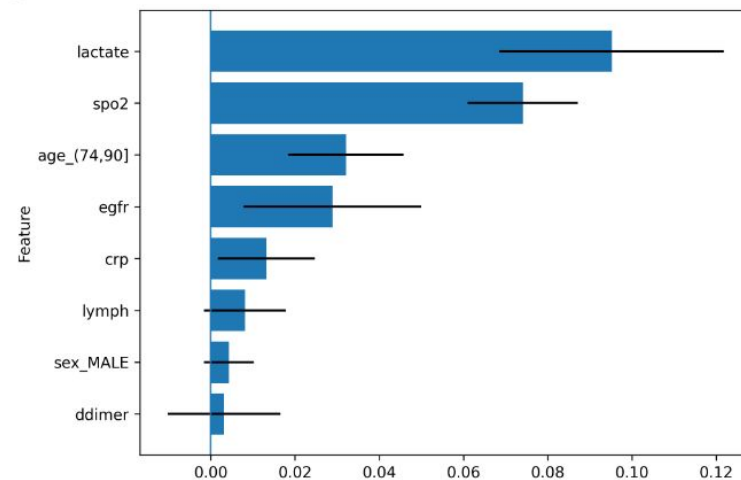
Validation-set permutation importance for the three clinical models

Permutation importance of clinical input features

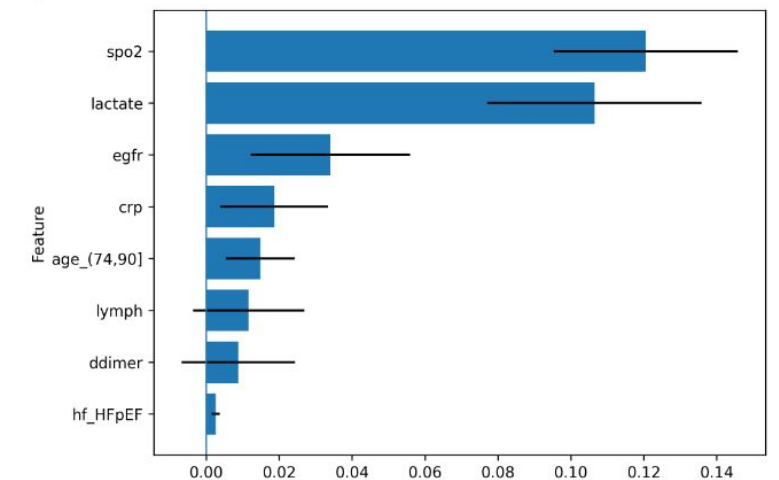
A Logistic Regression



B XGBoost



C MLP



Mean decrease in Validation ROC-AUC

Bars show the mean decrease in validation ROC-AUC when each feature is permuted; black horizontal lines indicate the variability across permutation repeats. Within each panel, features are ordered by importance; the horizontal axis is the same quantity in all three panels, but note that its range differs between panels.

Consistently high importance

Lactate and SpO₂ ranked among the top predictors across all three models; age, eGFR, and CRP also ranked highly in multiple models.

Permutation importance reflects model dependence on a feature — not a causal effect.

Full Performance Table and Statistical Comparisons

Model	ROC-AUC	95% CI	PR-AUC	Brier score	ECE
CXR ResNet18	0.907	0.836–0.963	0.623	0.102	0.152
Clinical Logistic Regression	0.941	0.887–0.981	0.800	0.057	0.041
Clinical XGBoost	0.936	0.876–0.979	0.718	0.068	0.090
Clinical MLP	0.880	0.768–0.961	0.689	0.169	0.309
Late Fusion	0.944	0.884–0.985	0.797	0.059	0.065

Paired DeLong tests with Holm correction

Family 1: Late Fusion vs clinical models		ΔROC-AUC	Holm-adjusted P	Family 2: Late Fusion vs component models		ΔROC-AUC	Holm-adjusted P
vs Clinical LR		0.003	0.950	vs Clinical LR		0.003	0.950
vs Clinical XGBoost		0.008	0.950	vs CXR ResNet18		0.037	0.092
vs Clinical MLP		0.065	0.062				

Late Fusion vs Clinical MLP was P = 0.021 before correction and 0.062 after Holm correction; Late Fusion vs CXR ResNet18 was P = 0.046 before correction and 0.092 after. A lack of statistical significance was not interpreted as evidence of equivalence.

Baseline Characteristics

Characteristic	Overall (N = 1,277)	Training (N = 1,021)	Validation (N = 128)	Test (N = 128)
In-hospital death, n (%)	169 (13.2%)	135 (13.2%)	17 (13.3%)	17 (13.3%)
Age 18–59 / 60–74 / 75–90, n	703 / 328 / 246	544 / 276 / 201	81 / 23 / 24	78 / 29 / 21
Female / Male / Missing, n	549 / 699 / 29	440 / 562 / 19	51 / 70 / 7	58 / 67 / 3
Heart failure No / HFpEF / HFrEF / Missing, n	968 / 42 / 26 / 241	775 / 36 / 21 / 189	98 / 6 / 2 / 22	95 / 0 / 3 / 30
SpO ₂ , median [IQR]	95 [92–97]	95 [92–97]	95 [92–97]	95 [92–98]
Lymphocyte count, median [IQR]	0.94 [0.68–1.32]	0.94 [0.68–1.31]	0.97 [0.72–1.41]	0.89 [0.69–1.19]
eGFR, median [IQR]	88.5 [56.0–108.0]	86.0 [57.0–107.0]	94.0 [55.8–109.2]	91.0 [54.5–111.0]
CRP, median [IQR]	8.45 [3.40–15.62]	8.40 [3.40–15.40]	8.25 [3.85–14.68]	8.60 [2.90–17.15]
D-dimer, median [IQR]	353.0 [214.2–642.8]	349.0 [209.0–655.0]	363.0 [235.5–598.0]	363.0 [242.5–631.8]
Lactate, median [IQR]	1.50 [1.10–2.00]	1.50 [1.10–2.00]	1.50 [1.10–1.92]	1.40 [1.00–1.90]

Missingness reflects the original data before imputation. No P values were calculated for comparisons across data splits.

TRIPOD+AI — Reporting Items Clarified in the Revision



Prediction time point



Eligibility criteria



Participant flow



Predictor selection



Missing-data handling



Patient-level data split



Separation of Training / Validation / Test roles



Hyperparameter and model-selection reporting



ROC-AUC with 95% confidence intervals



Calibration assessment



Limitations



Data and code availability

The study is reported in accordance with the TRIPOD+AI reporting guideline for clinical prediction model studies.

Full Paired DeLong Comparisons

ROC-AUCs were compared using paired DeLong tests. Multiple comparisons were adjusted using the Holm method.

$$\Delta\text{ROC-AUC} = \text{Late Fusion} - \text{comparator}$$

A. Comparison with clinical models — Holm family of three comparisons

Comparison	$\Delta\text{ROC-AUC}$	Unadjusted P	Holm-adjusted P	Interpretation
Late Fusion vs Clinical LR	+0.003	—	0.950	Not significant
Late Fusion vs Clinical XGBoost	+0.008	—	0.950	Not significant
Late Fusion vs Clinical MLP	+0.065	0.021	0.062	Not significant after Holm correction

B. Comparison with component models — separate Holm family of two comparisons

Comparison	$\Delta\text{ROC-AUC}$	Unadjusted P	Holm-adjusted P	Interpretation
Late Fusion vs Clinical LR	+0.003	—	—	Adjusted P not reported
Late Fusion vs CXR ResNet18	+0.037	0.046	0.092	Not significant after Holm correction

Late Fusion did not show a statistically significant improvement in ROC-AUC over any clinical model or either component model after Holm correction.

— Value not reported in the manuscript. The unadjusted P values for the comparisons with Clinical LR and Clinical XGBoost, and the family-B adjusted P value for Clinical LR, were not reported and have not been reconstructed here.

The two Holm families were adjusted separately; the four adjusted P values do not come from a single four-comparison family.

A non-significant difference should not be interpreted as equivalence.